



HK IMO Training 2023–2024
Problem Set 19
For 2 March 2024



Problem 73. There are $n \geq 2$ cities in a country. There is a road between every pair of cities, and there is a cost to use each road. Suppose the costs of using these roads are $1, 2, 3, \dots, \frac{n(n-1)}{2}$ respectively in some order.

Two people X and Y wish to travel all the cities in the following manner. X begins the journey at any city. Every time he chooses the road with the smallest cost to a city which he has not visited before, until he has visited all n cities. Y begins the journey at any city (not necessarily the same city as X). Every time he chooses the road with the largest cost to a city which he has not visited before, until he has visited all n cities. Is it possible to assign the costs to the roads in a certain way such that X and Y can travel all the cities with the same total cost?

Problem 74. Let a, b, c be given positive real numbers. Find the maximum value of

$$|ax + by - cz| + |ax - by + cz| + |-ax + by + cz|$$

where $x, y, z \in [0, 1]$.

Problem 75. Let AB be a diameter of a circle Γ with centre O . A chord CD of Γ is perpendicular to AB . Let M be the midpoint of CO . The line AM meets Γ again at E . The line BC meets AE and DE at P and Q respectively. The circumcircle of $\triangle DPQ$ meets Γ again at S . Prove that $PS = PB$.

Problem 76. Let $p \equiv 3 \pmod{4}$ be a prime number. Suppose θ is an angle such that $\tan \theta \in \mathbb{Q}$. Prove that $\tan((p+1)\theta)$ is a rational number $\frac{a}{b}$ (where $(a, b) = 1$) such that $p \mid a$.